

ECE4010J Probabilistic Methods in Engineering (Summer 2026)

Assignment 1

Date Due: See canvas

This assignment has a total of **(200 points)**.

Note: Unless specified otherwise, you must show the details of your work via logical reasoning for each exercise. Simply writing a final result (whether correct or not) will receive **0 point**.

Exercise 1.1 (30 pts) Let X and Y be independent random variables with $\mathbb{E}[X] = 3$, $\mathbb{E}[X^2] = 25$, $\mathbb{E}[Y] = 10$ and $\mathbb{E}[Y^2] = 164$.

- (i) (5 pts) Find $\mathbb{E}[3X - Y - 8]$
- (ii) (5 pts) Find $\text{Var } X$.
- (iii) (5 pts) Find $\text{Var } Y$.
- (iv) (5 pts) Find $\text{Var}[3X + Y - 8]$.
- (v) (5 pts) Find $\mathbb{E}[(X - 3)/4]$ and $\text{Var}[(X - 3)/4]$.
- (vi) (5 pts) Find $\mathbb{E}[(Y - 10)/8]$ and $\text{Var}[(Y - 10)/8]$.

Exercise 1.2 (50 pts) The zinc-phosphate coating on the threads of steel tubes used in oil and gas wells is critical to their performance. To monitor the coating process, an uncoated metal sample with known outside area is weighed and treated along with the lot of tubing. This sample is then stripped and reweighed. From this it is possible to determine whether or not the proper amount of coating was applied to the tubing. Assume that the probability that a given lot is unacceptable is 0.05. Let X denote the number of runs conducted to produce an unacceptable lot. Assume that the runs are independent in the sense that the outcome of one run has no effect on that of any other.

- (i) (10 pts) Verify that X is geometric. What is “success” in this experiment? What is the numerical value of θ ?
- (ii) (10 pts) What is the exact expression for the pmf for X ?
- (iii) (10 pts) What is the exact expression for the moment generating function for X ?
- (iv) (10 pts) What are the numerical values of $\mathbb{E}[X]$, $\mathbb{E}[X^2]$, and $\text{Var } X$?
- (v) (10 pts) Find the probability that the number of runs required to produce an unacceptable lot is at least 3.

Exercise 1.3 (70 pts) Consider the random variable X whose pmf is given by

$$p(x) = \frac{(x-3)^2}{5}, \quad x = 3, 4, 5$$

- (i) (10 pts) Verify that this function is a density for a discrete random variable.
- (ii) (10 pts) Find $\mathbb{E}[X]$ directly. That is, evaluate $\sum_x xp(x)$.
- (iii) (10 pts) Find $m_X(t)$, the moment generating function for X .
- (iv) (10 pts) Use the moment generating function to find $\mathbb{E}[X]$.
- (v) (10 pts) Find $\mathbb{E}[X^2]$ directly. That is, evaluate $\sum_x x^2p(x)$.
- (vi) (10 pts) Use the moment generating function to find $\mathbb{E}[X^2]$.
- (vii) (10 pts) Find σ^2 and σ .

Exercise 1.4 (20 pts) A discrete random variable has moment generating function

$$m_X(t) = e^{2(e^t - 1)}$$

- (i) (10 pts) Find $\mathbb{E}[X]$ and $\mathbb{E}[X^2]$.
- (ii) (10 pts) Find σ^2 and σ .

Exercise 1.5 (30 pts) (Discrete uniform distribution) A discrete random variable is said to be uniformly distributed if it assumes a finite number of values with each value occurring with the same probability. If we consider the generation of a single random digit, then Y , the number generated, is uniformly distributed with each possible digit occurring with probability $1/10$. In general, the pmf for a uniformly distributed random variable is given by

$$p(x) = 1/n, \quad x = x_1, x_2, \dots, x_n, \quad n \text{ a positive integer}$$

- (i) (10 pts) Find the moment generating function for a discrete uniform random variable.
- (ii) (10 pts) Use the moment generating function to find $\mathbb{E}[X]$, $\mathbb{E}[X^2]$, and $\text{Var } X$.
- (iii) (10 pts) Find the mean and variance for the random variable Y , the number obtained when a random digit generator is activated once. *hint:* The sum of the first n positive integers is $n(n+1)/2$; the sum of the squares of the first n positive integers is $n(n+1)(2n+1)/6$.